Dinh Thai Tuan

+84-386-010-391 | dinhthaituan48@gmail.com
SisypheanHUS | SisypheanHUS
Hanoi, Vietnam

Objective

Applied Mathematics graduate with experience in quantitative research, AI/ML, software development, and interdisciplinary academic research. Seeking opportunities in quantitative research, AI/ML, and software engineering, leveraging an Applied Mathematics background and experience across industry and academic research.

Education

- **Vietnam National University – University of Science** *Aug. 2022 – Sep. 2026*
  Bachelor of Science in Applied Mathematics Hanoi, Vietnam
  - GPA: 2.89/4
  - Main courses: Differential Equations, Calculus, PDEs, Probability Theory, Math Modelling, Statistics.
  - Thesis: Raman Spectroscopy for Methanol Detection
  - Advisor: Dr. Vu Duong – Institute of Physics, VAST
- **Nguyen Duy Trinh High School** *2019 – 2022*
  Secondary Education Nghi Loc, Nghe An, Vietnam

Experience

- **BiPlus Vietnam** *Nov. 2026 – Present*
  Java Developer – SAP Commerce Cloud (Hybris) Hanoi, Vietnam
  - Supporting enterprise e-commerce operations on SAP Commerce Cloud (Hybris), including order flows, catalogue, pricing, promotions, inventory, payment, and delivery.
  - Investigating technical issues through log analysis, configuration checks, SQL queries, and defect reproduction; managing technical tickets and escalations in Jira according to SLA requirements.
  - Supporting incident handling, Business Validation Testing (BVT), release validation, and technical documentation in collaboration with QA, DevOps, and client-side technical teams.
- **VinSmart Future** *Apr. – Jul. 2026*
  AI Engineer OJT Hanoi, Vietnam
  - Studied and evaluated machine learning approaches for fraud detection, including XGBoost and Random Forest, using the PaySim dataset.
- **Kaopiz** *Feb. – Mar. 2026*
  AI Engineer Intern (Part-time) Hanoi, Vietnam
  - Researched and evaluated AI-driven workflow automation tools, including OpenClaw for Jira work logging, web crawling, and multi-source data aggregation.
  - Evaluated Mem0 for chatbot memory and compared it with vector databases, retrieval-augmented generation (RAG), and alternative long-term memory architectures.
- **Vietnam Institute for Advanced Study in Mathematics** *Aug. – Nov. 2025*
  Summer Research Student Hanoi, Vietnam
  - Conducted research in epidemic modeling and actuarial science under the supervision of Dr. Hoang Manh Tuan.
  - Contributed to a manuscript on a modified SIS model for health insurance pricing.
- **Group of Mathematical Biology, Nara Women's University** *Aug. 2025*
  Summer Research Student Nara, Japan
  - Participated in a summer research program on dynamical systems in population biology.
  - Developed and presented an SIR modeling project in R.
- **Institute of Physics, Vietnam Academy of Science and Technology** *Jul. 2025 – Present*
  Research Intern Hanoi, Vietnam
  - Developed deep learning approaches for quantifying methanol concentration in ethanol–methanol mixtures using Raman spectroscopy.
  - Proposed and evaluated 1D and 2D CNN architectures for classifying methanol and ethanol from Raman spectral data.
  - First author of three conference papers presented at ICAEP-9 (2025) and ICPA-NCOS 14 / FEE 2026, including one oral and one poster presentation.
- **WorldQuant BRAIN** *Apr. 2024 – Present*
  Research Consultant Hanoi, Remote
  - Submitted 1,800+ alphas across global equity markets using Fast Expression Language and Python; built automated pipelines for template mining, correlation-based selection, and Power Pool allocation.
  - Achieved 4 Grandmaster and 2 Master rankings in the Genius Program, placing among the top 2% and top 8% tiers, respectively.
  - Selected as the sole delegate from Vietnam among 12 participants worldwide for the IQC 2025 finals in Singapore.

Skills

- **Programming Languages:** Python, Java, R, SQL, LaTeX
- **Frameworks & Tools:** PyTorch, TensorFlow, Scikit-learn, Git, Docker, Jira
- **Knowledge Areas:** Machine Learning, Deep Learning, Optimization, Factor Investing, Portfolio Management, E-commerce Systems

Publications

- **Dinh, T. T.**, N. T. Hieu, A. I. Vodchits, V. A. Orlovich, P. H. Minh, & V. Duong (2026). Raman spectroscopy identification of chemicals using a closed-form Bayesian marginal likelihood. *FEE Conference 2026*, oral presentation.
- **Dinh, T. T.**, N. T. Hieu, P. H. Minh, & V. Duong (2026). Raman peak matching for component detection in compounds. *ICPA-NCOS 14*.
- **Dinh, T. T.** & V. Duong (2025). Raman spectroscopy for methanol detection: 1D vs. 2D CNN performance evaluation. *Proc. ICAEP-9*.

Certifications, Honors and Awards

- **1st prize,** International Quant Championship 2024 – University Round (37,000+ participants) *2024*
- **9th place,** International Quant Championship 2024 – National/Regional Round *2024*

Academic Activities

- VIASM Summer School on Mathematical Physics 2026, HCMUS *Aug. 2026*
- Summer School on Mathematical Biology 2025, VIASM *Jul.–Aug. 2025*
- Samsung Innovation Campus, Artificial Intelligence Course *2024*

Additional Information

Languages: Vietnamese (Native), English (TOEIC 890 – L/R 460/430)
Interests: Quantitative Research, Machine Learning, Factor Investing, Mathematical Modelling